but depends on several parameters related to the organism, the host and their interaction.

Microorganisms first enter the body, survive, multiply and elaborate many factors and produce the disease.

Adhesion: The initial event in the pathogenesis of many infections is the attachment of the bacteria to body surfaces. Adhesions may occur as organized structures, such as fimbriae and pili. Adhesions serve as virulence factors.

Capsule: It is an envelope or slime layer surrounding the cell wall of certain microorganisms. Capsule plays important roles in immune evasion as it inhibit's phagocytosis, as well as protecting the bacteria while outside the host.

Toxins: Toxins are specific chemical products of microbes, plants and some animals that are poisonous to other organisms. Toxigenicty is the power to produce toxins.

A toxin is named according to specific target of action: Neurotoxin acts on the nervous system. Enterotoxin acts on the intestine, Haemotoxin lyses red blood cells, and Nephrotoxins damages the kidneys.

A toxin molecule secreted by a living bacterial cell into the infected tissue is an **exotoxin**. A toxin that is not actively secreted but is shed from the outer membrane is an **endotoxin**. the difference between exotoxin and endotoxin were given in Table 12.1.

Production of enzymes

Some enzymes like proteases, DNAases and phospholipases are produced and they help in destruction of the cell structure and to hydrolyse host tissues.

Antigenic variation

Microorganisms evade the host immune responses by changing their surface antigens. Antigenic drift and antigenic shift are common in influenza viruses. The distinction between the commensal and the organism associated with disease.

12.1.5 Diagnostic Cycle

Specific diagnosis is important for better patient care, use of appropriate antibiotics

Table 12.1: Differences between endotoxin and exotoxin

Exotoxins	Endotoxins
Heat labile proteins, secreted by certain species of bacteria and diffuse readily into the surrounding medium	Heat stable polysaccharide proteins, lipid complex which form an integral part of the cellwall of Gram negative bacteria
Proteins with a strong specificity to a target cell and extremely powerful sometimes deadly	A Lipopolysaccharide (LPS), which is part of the outermembrane of gram negative cell walls
Highly immunogenic	Less immunogenic
Toxoids can be made by treating toxins with formalin	Toxoids cannot be made
Produced mainly by Gram positive bacteria but also by some Gram negative bacteria	Produced by Gram negative bacteria

TN_GOVT_XI_Micro_Biology_CH12.indd 187 16-02-2019 10.05.39 AM